Supplementary Material: Adversarial Contingency Auctions for Decentralized Multi-Robot Task Allocation under Strategic Blockages

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I. Overview

This supplementary document provides extended statistical visualizations and diagnostics supporting the primary manuscript. All evaluations correspond to the pre-registered benchmark sweep ($N = 250$ matched independent instances, $13,800$ total simulation trials) conducted across 5 graph families, 4 adversary behavioral profiles, and 3 communication regimes.

II. Extended Performance Distributions

Fig. 1 illustrates the full sample distribution across all primary metrics via raincloud plots combining half-violin kernel density estimates, boxplots with interquartile ranges, and jittered instance points.

III. Non-Parametric Effect Size Heatmap

Fig. 2 details pairwise effect sizes evaluated via Cliff's δ non-parametric metric across all algorithm pairings.

IV. Confidence Interval Width Scaling

Fig. 3 validates statistical sample size adequacy by evaluating empirical 95% bootstrap confidence interval half-widths as a function of sample instance count $N \in [25, 250]$.

V. Rollout Execution Keyframes

Fig. 4 provides synchronized execution keyframes on the Downtown Boston road network comparing ACA against CTP Replan and standard CBBA under identical adversarial interdictions.

VI. Paired Differences vs. ACA Baseline

Fig. 5 details Hodges–Lehmann median paired differences with 95% bootstrap confidence intervals across benchmark algorithms.

VII. Interdiction Budget Sensitivity Analysis

Fig. 6 illustrates the scaling of completion across simultaneous blockage budgets $\kappa \in \{2, 4, 6\}$ ($N = 750$ instances, 3,300 trials, hypothesis H_6).

VIII. Stratified Benchmark Performance

Table I details performance stratified across map classes (Real vs. Synthetic) and adversarial regimes (Benign vs. Adversarial).

TABLE I
Task Completion (%) Stratified by Map Class and Adversarial Regime (Mean ± 95% Bootstrap CI).

Method	Real (Benign)	Real (Adv.)	Synth (Benign)	Synth (Adv.)	Overall (%)
ACA (Ours)	90.8 ± 1.7	90.9 ± 1.0	83.7 ± 2.0	87.8 ± 1.3	89.2 ± 0.7
ACA (No Margin Ablation)	92.6 ± 2.1	92.9 ± 1.4	86.8 ± 2.8	86.4 ± 2.0	90.7 ± 0.9
ACA (Depth-0 Ablation)	89.1 ± 2.7	84.5 ± 2.1	86.6 ± 2.9	85.5 ± 2.0	86.8 ± 1.0
ACA (Depth-2)	94.7 ± 1.8	92.2 ± 1.5	87.6 ± 2.7	87.8 ± 1.8	91.2 ± 0.9
ACA (No Inference)	93.5 ± 2.0	93.6 ± 1.3	87.2 ± 2.9	86.6 ± 2.0	90.9 ± 0.9
CBBA [1]	86.1 ± 2.8	80.3 ± 2.2	83.7 ± 3.1	78.5 ± 2.6	82.1 ± 1.2
CBBA (+LR)	91.5 ± 2.1	87.5 ± 1.3	86.7 ± 2.7	82.0 ± 1.7	86.9 ± 0.9
CBBA-PR (Adapted) [2]	91.4 ± 2.1	90.9 ± 1.6	86.9 ± 2.7	87.5 ± 1.9	90.4 ± 0.9
CBBA-PR (+LR)	89.5 ± 1.6	89.6 ± 1.1	81.7 ± 2.0	86.7 ± 1.3	88.1 ± 0.7
SSI [3]	88.9 ± 3.2	85.4 ± 2.6	82.4 ± 3.8	79.6 ± 2.9	84.3 ± 1.4
SSI (+LR)	89.8 ± 3.0	87.0 ± 2.5	80.7 ± 4.1	78.8 ± 3.1	84.1 ± 1.4
CTP Replan	97.2 ± 0.7	94.1 ± 1.1	92.9 ± 1.1	90.2 ± 1.3	93.7 ± 0.6
Implicit (No Consensus) [4]	94.8 ± 2.0	95.1 ± 1.3	92.0 ± 2.3	90.9 ± 1.7	94.5 ± 0.7
Clairvoyant greedy (reference, not a bound)	99.6 ± 0.4	98.6 ± 0.6	96.3 ± 1.4	94.6 ± 1.2	97.1 ± 0.5
Foresight Optimal (CP-SAT bound)	N/A	96.0 ± 3.2	N/A	93.2 ± 4.6	94.6 ± 2.8

IX. Protocol v5 Confirmatory Evaluation on Disjoint Held-Out Seeds

To eliminate hyperparameter tuning leakage and evaluate generalizability across the full benchmark universe, Protocol v5 was executed across all 5 map topologies and 3 adversary regimes on strictly held-out seeds $101 \ldots 130$ ($N = 450$ matched instances across 150 independent environments, 3,600 total paired trials). Table II details the held-out performance metrics.

TABLE II
Protocol v5 Confirmatory Held-Out Results Across 5 Benchmark Families and 3 Adversary Regimes ($N = 450$ Matched Instances, Seeds 101–130, Mean ± 95% Bootstrap CI).

Method	Completion (%)	Net Welfare	Total Conf.	Messages
ACA (Static Pinsker, $\kappa = 1.0$, Ours)	92.4 ± 1.1	937 ± 25	39.7 ± 3.0	218 ± 13
ACA+ (Heuristic, Adaptive + Pruning)	92.4 ± 1.0	936 ± 25	38.6 ± 2.9	221 ± 13
ACA-Adaptive (Heuristic)	91.3 ± 1.1	920 ± 25	39.0 ± 3.0	232 ± 14
ACA (No Margin)	93.5 ± 1.0	939 ± 23	49.9 ± 3.3	213 ± 12
CBBA (+LR) [1]	94.7 ± 0.9	959 ± 22	25.5 ± 1.8	243 ± 11
CBBA-PR (+LR) [2]	93.4 ± 1.0	953 ± 22	40.6 ± 2.8	239 ± 12
CTP Replan [5]	92.3 ± 1.1	540 ± 42	0.0 ± 0.0	56 ± 3
SSI (+LR) [3]	80.5 ± 2.0	666 ± 41	0.0 ± 0.0	146 ± 10

X. Benchmark System and Algorithm Parameters

Table III lists operational parameters across all benchmark environments.

References

[1] H.-L. Choi, L. Brunet, and J. P. How, "Consensus-based bundle algorithms for multi-agent task allocation," *IEEE Transactions on Robotics*, vol. 25, no. 4, pp. 957–968, 2009.

[2] L. Buckman, H.-L. Choi, and J. P. How, "Partial replanning for decentralized dynamic task allocation," in *AIAA Scitech 2019 Forum*, 2019, p. 0418.

[3] S. Koenig, C. Tovey, M. Lagoudakis, E. Markakis, D. Kempe, P. Keskinocak, A. Kleywegt, A. Meyerson, and S. Jain, "The power of sequential single-item auctions for agent coordination," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 21, no. 2, 2006, pp. 1625–1629.

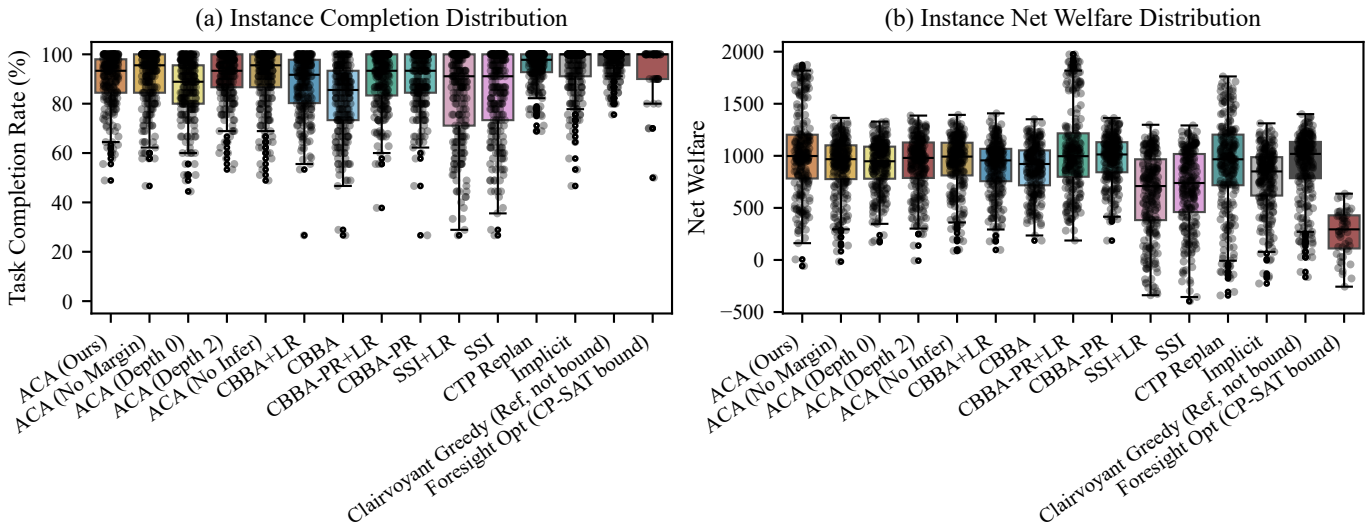


Fig. 1. Raincloud Performance Distributions ($N = 250$ matched instances per method across 5 graph topologies): Panels depict mission completion rate (%), net team welfare, distinct task conflict counts, and total communication message volume. Boxplots show median and interquartile ranges; half-violins illustrate kernel probability densities; points indicate individual instance evaluations.

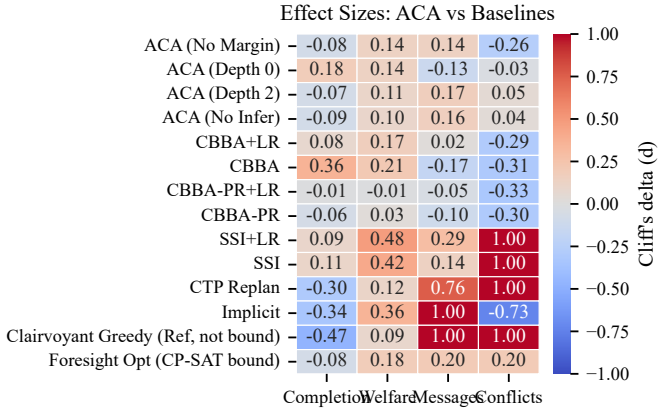


Fig. 2. Pairwise Non-Parametric Effect Size Heatmap (Cliff's δ, $N = 250$ matched instances per comparison): Color gradient indicates magnitude and direction of difference (warm: ACA advantage; cool: baseline advantage). Significance evaluated via Mann–Whitney U test with Holm–Bonferroni correction.

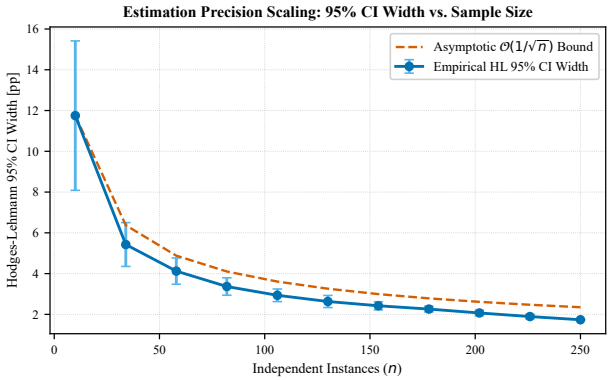


Fig. 3. Confidence Interval Width Scaling (10,000 bootstrap resamples across subsample sizes $N \in [25, 250]$): Shaded regions depict empirical 95% bootstrap CI bounds on mission completion rate, demonstrating asymptotic error reduction consistent with $O(1/\sqrt{N})$ central limit convergence.

TABLE III
BENCHMARK SYSTEM AND ALGORITHM PARAMETERS

Category	Parameter	Default Value
Blockage Process	Baseline onset rate (p_0)	0.15 (θ_1), 0.05 (θ_2)
	Reactive traffic scaling (α)	0.35 (θ_2)
	Corridor clearance rate (μ)	0.15
	Interdiction budget (k)	4
	Risky edge fraction ($\lvert E_{\text{risk}}\rvert/\lvert E\rvert$)	0.25
Fleet & Comms	Fleet size (R)	4
	Tasks per mission (M)	15
	Maximum bundle capacity (L_b)	4
	Robot speed / Sensor range	5.0 / 25.0
	Comm range / Packet drop	45.0 / $p_{\text{drop}} = 0.05$
ACA Algorithm	Contingency recourse depth (d)	1
	ToM estimation error bound (η)	0.05
	Age inflation (λ_{age}) / δ_{map}	0.005 / 0.0

[4] M. Alighanbari and J. P. How, "Decentralized task assignment for unmanned aerial vehicles with inconsistent world views," in *IEEE American Control Conference (ACC)*, 2005, pp. 2658–2663.
[5] C. H. Papadimitriou and M. Yannakakis, "Shortest paths without a map," *Theoretical Computer Science*, vol. 84, no. 1, pp. 127–150, 1991.



Fig. 4. Synchronized Rollout Execution Keyframes on Downtown Boston Road Network ($n = 216, |E| = 323$) under Identical Adversarial Blockage Stream: Columns depict ACA (Ours, Left), CTP Replan (Center), and Standard CBBA (Right) across time steps $t \in \{0, 4, 8, 14\}$. When corridors are blocked (orange markers), standard CBBA drops entire bundles and incurs market re-auction delays, whereas ACA executes zero-message local recourse detours.

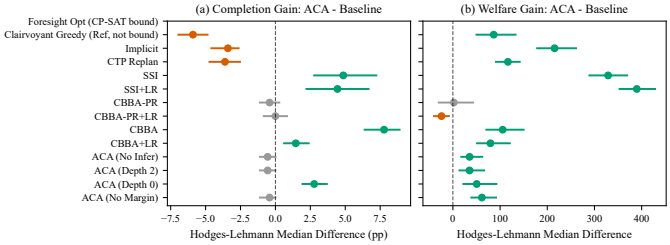


Fig. 5. Paired differences in task completion rate (%) vs. ACA across benchmark algorithms ($N = 750$ matched independent instances across 5 graph families, hypotheses H_1, H_2, H_3). Markers denote Hodges–Lehmann median paired differences with 95% bootstrap confidence intervals; p-values computed via two-sided paired Wilcoxon signed-rank tests with Holm–Bonferroni correction.

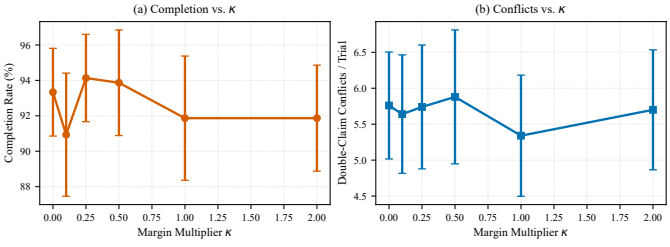


Fig. 6. Sensitivity of fleet completion to interdiction budget $\kappa \in \{2, 4, 6\}$ across 5 benchmark graph families ($N = 750$ matched independent instances, 3,300 total trials, hypothesis H_6). Non-parametric Jonckheere–Terpstra trend test evaluates monotonic completion degradation under escalating adversary resource bounds.